

PROJECT PROGRESS UPDATE GUIDE

Standard Operating Procedures for Intern Projects
Version 1.0 | Team-Based Execution | Mandatory GitHub Compliance

Applic Coders Software Solutions - Internship Program (3 Months)

To ensure every intern successfully completes the internship, each month consists of 4 Guidance Sessions and 4 Project Update Meetings. Guidance sessions focus on learning, mentoring, and resolving technical issues, while Project Update Meetings are used to review progress, evaluate performance, and assign upcoming tasks.

Team Project Guidelines – Complete 2 out of 3 Projects (Week-wise Implementation Plan)

- Program Start Date: 10th of every month
- Program Duration: 3 Months
 - Month 1: Project 1
 - Month 2: Project 2
 - Month 3: Project 3
- Implementation Plan: 4-week milestone-based execution for each project
- GitHub Repository: Mandatory for all technical domains (except Figma/UI Design projects)
- Team Size: To be determined by Applic Coders Software Solutions
- Assigned Role: Python Developer
- Project Assignment Authority: Pyintel Computing Lab

AegisLogix Cloud-Auditor (IT Operations Domain)

Python Developer Technical Assessment – Phase-wise Development Plan

The project is expected to be implemented over 3 months, following an enterprise software development lifecycle. Each phase represents the minimum expected completion percentage and deliverables.

Month 1 – Foundation Phase (35% Completion)

Objective

Develop the core application architecture and implement the fundamental modules.

Deliverables

Project Setup

- Django project architecture
- PostgreSQL database configuration
- Virtual environment setup
- Environment variable configuration (.env)
- Git repository setup

Authentication & Security

- Django Authentication
- Role-Based Access Control (Admin/User)
- Password hashing
- Login & Logout
- Session Management
- CSRF Protection
- Input Validation
- Secure File Upload Validation

Dashboard (Basic)

- Responsive UI using HTML5, CSS3, Bootstrap
- Dashboard Layout
- Login Dashboard
- User Profile
- Navigation

Log Upload Module

- Upload server log files
- Validate file size and type
- Store upload history
- Duplicate file detection using SHA-256 hash

Database

- Design normalized PostgreSQL schema
- Models and migrations
- Basic CRUD operations

Python Requirements

- Object-Oriented Programming
- Modular coding standards
- Custom Exceptions
- Logging Framework
- Type Hinting
- PEP-8 compliance

Month 1 Completion Target

35% Project Completion

Month 2 – Security & Cloud Integration (75% Completion)

Objective

Implement enterprise security features, advanced Python processing, and AWS cloud integration.

Deliverables

Advanced Python Processing

- Multi-threaded log parsing
- ThreadPoolExecutor / ProcessPoolExecutor
- Custom decorators
- Context Managers
- Generators
- Dataclasses
- SOLID Principles
- Factory Design Pattern
- Strategy Pattern

Threat Detection Engine

Detect:

- Brute Force Attacks
- SQL Injection
- Cross-Site Scripting (XSS)

- Command Injection
- Directory Traversal
- Port Scanning
- Unauthorized IP Access
- Privilege Escalation
- Denial of Service (DoS)
- Abnormal Login Attempts
- Suspicious User Activity

Security Enhancements

- Secure HTTP Headers
- Content Security Policy (CSP)
- Secure Cookies
- Password Policy Enforcement
- Login Attempt Limiting
- Account Lockout
- Audit Logs
- IP Tracking
- Rate Limiting
- JWT Authentication (API)
- CORS Configuration
- Encryption of Sensitive Data

AWS Integration

- AWS S3 Integration using Boto3
- Automatic Archive
- Lifecycle Policies
- Pre-Signed URLs
- Server-Side Encryption
- Archive Metadata

PostgreSQL Optimization

- Indexing
- Query Optimization
- Aggregation
- Bulk Insert
- Transactions
- `select_related()`
- `prefetch_related()`

- Dashboard
- Threat Statistics
- Compliance Score
- Real-Time Alerts
- Top Attack Sources

Monthly Reports

Month 2 Completion Target

75% Project Completion

Month 3 – Client Requirements & Production Readiness (100% Completion)

Objective

Complete all remaining client-specific requirements, optimize the application, and prepare it for deployment.

Deliverables

Client-Specific Enhancements

- Implement all pending client-requested features.
- Modify workflows based on client feedback.
- Develop additional reports and dashboards.
- Add custom business logic as required.

Performance Optimization

- Database tuning
- Query optimization
- Code profiling
- Memory optimization
- Large file handling
- Background task optimization

API Development

- Django REST Framework APIs
- Swagger/OpenAPI Documentation
- Postman Collection
- API Versioning

Testing

- Unit Testing
- Integration Testing
- Security Testing
- Load Testing
- Regression Testing
- User Acceptance Testing (UAT)

DevOps & Deployment

- Docker & Docker Compose
- Nginx Configuration
- Gunicorn Setup
- CI/CD Pipeline (GitHub Actions/Jenkins)
- Production Environment Configuration

Documentation

- Technical Documentation
- Architecture Diagram
- Database Design
- Deployment Guide
- User Manual
- API Documentation
- Installation Guide

Final Deliverables

- Complete source code
- Production-ready application
- Database scripts
- Test cases
- Documentation
- AWS deployment configuration
- Client handover package

Month 3 Completion Target

100% Project Completion

Final Evaluation Criteria

- Python & Advanced Python Programming
- Django Framework Expertise
- PostgreSQL Design & Optimization
- AWS S3 Integration
- Secure Coding Practices
- Application Performance
- Code Quality & Maintainability
- API Development
- Testing Coverage
- Documentation
- Scalability
- Production Readiness

Note:

The candidate is expected to complete 35% of the assigned project by the end of Month 1, 75% by the end of Month 2, and the remaining 25% by the end of Month 3, including all client-specific enhancements, testing, optimization, deployment, and documentation.

Upon successful completion of the project, an interview will be scheduled based on the candidate's overall performance throughout the project duration. Candidates who do not meet the required performance standards or fail to demonstrate satisfactory progress will not be eligible for the interview process.